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Multi-purpose traction inverter bus bar system

Abstract

A common lubrication and cooling system includes a lubricant-supported electric motor including a stator and a rotor defining a gap therebetween, with a liquid coolant disposed in the gap for supporting the rotor while allowing the rotor to rotate relative to the stator. The common lubrication and cooling system also includes a liquid-cooled inverter. The liquid-cooled inverter includes a plurality of switch devices configured to supply an alternating current (AC) power to the lubricant-supported electric motor for driving the rotor to rotate. The liquid-cooled inverter also includes a first heatsink mechanically connected to the plurality of switch devices. The liquid-cooled inverter also includes an inverter passageway configured transmit the liquid coolant between the lubricant-supported electric motor and into fluid communication with the first heatsink for transmitting heat away from the first heatsink.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This utility application claims the benefit of U.S. Provisional Application No. 63/274,154 filed Nov. 1, 2021. The entire disclosure of the above application is incorporated herein by reference.

FIELD OF THE DISCLOSURE
(1) The present disclosure relates generally to a lubricant-supported electric motor and a liquid-cooled inverter. More specifically, the present disclosure relates to a multi-purpose traction inverter bus bar system used in association with a liquid-cooled inverter that allows the liquid-cooled inverter and the lubricant-supported electric motor to share a common lubricant/coolant fluid.
BACKGROUND OF THE INVENTION
(2) This section provides a general summary of background information and the comments and examples provided in this section are not necessarily prior art to the present disclosure.
(3) Various drivelines in automotive, truck and certain off-highway applications take power from a central prime mover and distribute the power to the wheels using mechanical devices such as transmissions, transaxles, propeller shafts, and live axles. These configurations work well when the prime mover can be bulky or heavy, such as, for example, various internal combustion engines (“ICE”). However, more attention is being directed towards alternative arrangements of prime

movers that provide improved environmental performance, eliminate mechanical driveline components, and result in a lighter-weight vehicle with more space for passengers and payload.

(4) "On wheel", "in-wheel" or "near-wheel" motor configurations are one alternative arrangement for the traditional ICE prime mover that distributes the prime mover function to each or some of the plurality of wheels via one or more motors disposed on, within, or proximate to the plurality of wheels. For example, in one instance, a traction motor, using a central shaft though a rotor and rolling element bearings to support the rotor, can be utilized as the "on wheel", "in wheel" or "near wheel" motor configuration. In another instance, a lubricant-supported electric motor can be utilized as the "on wheel", "in wheel" or "near wheel" motor configuration. While each of these motor configurations result in a smaller size and lighter weight arrangement as compared to the prime movers based on the internal combustion engine, they each have certain drawbacks and disadvantages.

(5) One aspect of electric drive motors that adds to their cost and complexity is the requirement for a variety of fluids used for different functions of the electric drive motor and associated systems. For example, a wheel end electric system often includes a water-glycol cooling fluid for the electric motor and separate cooling fluids for other components (e.g., the liquid-cooled inverter) of the electric drive system. Each of these fluids require separate storage compartments and distribution channels, as well as systems for distributing or cycling the fluids to the desired locations within the systems. In the water-glycol cooled traction motors, the cooling fluid also does not touch the moving motor parts, such as the bearing surfaces, and thus cannot cool these components and is unable to support the rotor relative to the stator, such as is the case with lubricant-supported electric motors. In addition, fluid passages defined by the electric motor are not narrow enough to block the flow of the coolant.

(6) Additionally, water-based coolants must be separated from hydrocarbon lubricated surfaces and from high voltage and low voltage electronics. A water-glycol based coolant coming into contact with electronics can lead to electrical shorts and substantial damage to the electrical components. Thus, using water-glycol coolants to cool electronics requires the use of heat exchangers, which are themselves costly, bulky and heavy. Accordingly, most inverters require that the electronic components are separated by an aluminum plate from the coolant fluid, so that the fluid is never in contact with the power components. Put another way, prior art liquid-cooled inverters mount the devices to an aluminum plate (via a thin insulator) which is in contact with the fluid. If the water-glycol is the cooling fluid, the aluminum plate is grounded. If oil is the cooling fluid, the aluminum plate can be isolated from ground, but an additional cost is required because of the relatively high mass of the aluminum plate and power devices. As a result, the heatsink is usually grounded because of the mounting design.

(7) For these reasons, the present solutions to the problem of cooling liquid-cooled inverters with oil results in the use of a 2.sup.nd fluid (water-glycol) or an expensive mechanical design. Neither one of these solutions is advantageous as it leads to increased costs due to additional parts or cooling systems. Also, in most of the prior art applications, over-heating of the dc link capacitor is a major design difficulty because the outside surface of the package is not easily cooled. Thus, there remains a continuing need for improved systems to allow a shared cooling to be used for both a lubricant-supported electric motor as well as to a liquid-cooled inverter.

SUMMARY OF THE INVENTION

(8) The present disclosure provides a common lubrication and cooling system. The common lubrication and cooling system includes a lubricant-supported electric motor including a stator and a rotor defining a gap therebetween, with a liquid coolant disposed in the gap for supporting the rotor while allowing the rotor to rotate relative to the stator. The common lubrication and cooling system also includes a liquid-cooled inverter. The liquid-cooled inverter includes a plurality of switch devices configured to supply an alternating current (AC) power to the lubricant-supported electric motor for driving the rotor to rotate. The liquid-cooled inverter also includes a first heatsink

mechanically connected to the plurality of switch devices. The liquid-cooled inverter also includes an inverter passageway configured transmit the liquid coolant between the lubricant-supported electric motor and into fluid communication with the first heatsink for transmitting heat away from the first heatsink.
(9) The present disclosure also provides a liquid-cooled inverter. The liquid-cooled inverter includes a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween. The liquid-cooled inverter also includes a phase driver. The phase driver includes a printed circuit board (PCB), a high-side power switch configured to selectively conduct power between the DC positive conductor and an output conductor, and a low-side power switch configured to selectively conduct power between the DC negative conductor and the output conductor. At least one of the high-side power switch and the low-side power switch includes a plurality of switch devices disposed on the PCB. The liquid-cooled inverter also includes a first heatsink mechanically connected to the plurality of switch devices and configured to remove heat therefrom. The liquid-cooled inverter also includes a housing defining an inverter passageway configured to conduct a liquid coolant in thermal communication with the first heatsink for removing heat therefrom.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** shows a schematic view of a system including a lubricant-supported electric motor and a liquid-cooled inverter having a shared coolant/lubricant in accordance with an aspect subject disclosure;
(3) FIG. **2** shows an electrical schematic diagram showing a portion of the liquid-cooled inverter in accordance with an aspect of the present disclosure;
(4) FIG. **3** shows a perspective view of a printed circuit board assembly (PCBA) of the liquid-cooled inverter in accordance with an aspect of the present disclosure;
(5) FIG. **4** shows a top view of the PCBA of FIG. **3**, with a heat sink; and
(6) FIG. **5** shows a perspective view of the liquid-cooled inverter, including a stack of three of the PCBAs.
DETAILED DESCRIPTION OF THE ENABLING EMBODIMENTS
(7) The present disclosure is generally directed to a lubricant-supported electric motor and an electric component, namely a liquid-cooled inverter, having a shared lubricating and cooling system. More specifically, the system uses a common lubricant/coolant fluid, such as a dielectric oil, that is both disposed within a lubricant-supported electric motor to lubricant the electric motor and support the rotor relative to the stator, while also being used to cool electronic modules of the liquid-cooled inverter. Put another way, the lubricant-supported electric motor and the liquid-cooled inverter use the same liquid coolant. Oil is a very good liquid coolant to use for cooling high voltage components because of the oil's insulating properties. The oil may also help to minimize electromagnetic interference (EMI) by spacing apart current carrying devices from grounded conductors. Other advantages will be appreciated in view of the following more detailed description of the subject invention.
(8) FIG. **1** illustrates a common lubrication and cooling system **10** for a lubricant-supported electric motor **11** of the disclosure. As best illustrated in FIG. **1**, the lubricant-supported electric motor **11** includes a stator **12** and a rotor **14** extending along an axis A and movably disposed within the stator **12** to define a support chamber **16** or gap G therebetween. Although illustrated and described with the rotor **14** disposed within the stator **12**, the arrangement of these components can also be

reversed (i.e., with the rotor **14** disposed in surrounding relationship with the stator **12**) without departing from the scope of the subject disclosure. A lubricant/coolant fluid **18** is disposed in the support chamber **16** for supporting the rotor **14** within and relative to the stator **12**, allowing the rotor **14** to rotate relative to the stator **12** and lubricating and cooling these components. The lubricant/coolant fluid **18** acts as a buffer (e.g., suspension) between the rotor **14** and stator **12** minimizing or preventing contact therebetween. In other words, the lubricant/coolant fluid **18** prevents direct contact between the stator **12** and rotor **14** and provides a lubricant-supported electric motor **11** which is robust to shock and vibration loading due to the presence of the lubricant/coolant fluid **18** in the support chamber **16**. Additionally, and alternatively, a lubricant/coolant fluid that is substantially incompressible may be used in order to minimize the gap between the stator **12** and rotor **14**.

(9) As further illustrated in FIG. **1**, the rotor **14** is interconnected to a drive assembly **20** for coupling the lubricant-supported electric motor **11** to one of the plurality of wheels of a vehicle. For example, in one instance, the drive assembly **20** may include a planetary gear system. Alternatively, the drive assembly **20** may include one or more parallel axis gears. In either arrangement, the lubricant-supported electric motor **11** is arranged in an "on-wheel", "near-wheel" or "in-wheel" motor system in which the lubricant-supported electric motor **11** is disposed proximate to, on, or within the vehicle wheel. Although not expressly illustrated, accordingly to another aspect, the lubricant-supported electric motor **11** can be connected directly to the vehicle wheel, without the use of this drive assembly **20** to establish the "on-wheel", "near-wheel" or "in-wheel" electric motor arrangement. In any arrangement, the stator **12** and rotor **14** are configured to exert an electromagnetic force therebetween to convert electrical energy into mechanical energy, moving the rotor **14** and ultimately driving the wheel coupled to the lubricant-supported electric motor **11**. If present, the drive assembly **20** may provide one or more reduction ratios between the lubricant-supported electric motor **11** and the wheel in response to movement of the rotor **14**.

(10) As further illustrated FIG. **1**, the stator **12** defines a motor passageway **22** disposed in fluid communication with the support chamber **16** for introducing the lubricant/coolant fluid **18**. However, the motor passageway **22** could be provided on any other components of the lubricant-supported electric motor **11**, without departing from the subject disclosure. A liquid-cooled inverter **24** is disposed in electrical communication with the lubricant-supported electric motor **11** and defines an inverter passageway **26** disposed in fluid communication with the motor passageway **22** for allowing the lubricant/coolant fluid **18** to also pass through the liquid-cooled inverter **24** and over its electronic components. Thus, the lubricant/coolant fluid **18** used to lubricate and cool the lubricant-supported electric motor **11** is also used for cooling the liquid-cooled inverter **24**. The lubricant/coolant fluid **18** may be a liquid, such as a dielectric oil having a composition that acts as an electrical insulator, such that the lubricant/coolant fluid **18** will not conduct electricity, making the lubricant/coolant fluid **18** suitable for direct contact with electric components of the liquid-cooled inverter **24**. The dielectric oil also has good heat transfer properties, such that it may act well as a coolant for both the lubricant-supported electric motor **11** as well as the liquid-cooled inverter **24**. The dielectric oil is also incompressible, making it a good candidate for supporting the rotor **14** relative to the stator **12** in the lubricant-supported electric motor **11**. Finally, the dielectric oil also serves as a good lubricant for use within the lubricant-supported electric motor **11**. The liquid-cooled inverter **24** includes a number of electric components necessary to convert DC current into AC current, such as switches, transistors, semiconductors and the like.

(11) As illustrated in FIG. **1**, the lubricant/coolant fluid **18** is cycled or pumped through both the motor passageway **22** and the inverter passageway **26**, and into their respective components, as one continuous fluid communication line. For example, a pump **36** may be fluidly coupled to a sump or reservoir **38** of the lubricant/coolant fluid **18**, such that the lubricant/coolant fluid **18** is pumped from the reservoir **38**, through the motor passageways **22** and the inverter passageways **26** and into the support chamber **16** of the lubricant-supported electric motor **11**. In an alternative arrangement,

rotation of the rotor **14** relative to the stator **12** could act as a self-pump to pull the lubricant/coolant fluid **18** through the passageways **22**, **26** and into the support chamber **16**. Although not expressly illustrated, a further enhancement of the common lubrication and cooling system **10** includes that the reservoir **38** is designed with a low point where water present in the lubricant/coolant fluid **18** could collect. A diagnostic message to a driver of the vehicle could be sent indicating that a drain plug of the reservoir **38** needs to be opened to purge the tank of water.

(12) FIG. **2** shows an electrical schematic diagram of a portion of the liquid-cooled inverter **24**. The liquid-cooled inverter **24** includes a DC positive conductor **40***a* and a DC negative conductor **40***b* having a DC voltage therebetween. The liquid-cooled inverter **24** includes an A-phase driver **42***a* configured to generate an alternating current (AC) power on an A-phase output conductor **44***a* for supplying current to a corresponding stator winding of the lubricant-supported electric motor **11**. The liquid-cooled inverter **24** also includes a B-phase driver **42***b* configured to generate AC power on a B-phase output conductor **44***b* for supplying current to a corresponding stator winding of the lubricant-supported electric motor **11**. The liquid-cooled inverter **24** also includes a C-phase driver **42***c* configured to generate AC power on a C-phase output conductor **44***c* for supplying current to a corresponding stator winding of the lubricant-supported electric motor **11**. Each of the phase drivers **42***a*, **42***b*, **42***c* includes a high-side power switch **46***h* configured to selectively conduct current between the DC positive conductor **40***a* and a corresponding one of the output conductors **44***a*, **44***b*, **44***c*. Each of the phase drivers **42***a*, **42***b*, **42***c* also includes a low-side power switch **461** configured to selectively conduct current between the DC negative conductor **40***b* and a corresponding one of the output conductors **44***a*, **44***b*, **44***c*. Each of the power switches **46***h*, **461** is shown schematically as a single insulated gate bipolar transistor (IGBT) having a collector terminal C, an emitter terminal E, and a gate terminal G. Each of the power switches **46***h*, **461** is configured to selectively conduct current between the collector terminal C and the emitter terminal E in response to application of a control voltage to the gate terminal G. Alternatively or additionally, other types of power switching devices, such as field-effect transistors (FETs) or other types of junction devices, may be used for some or all of the power switches **46***h*, **461**. In some embodiments, one or more of the power switches **46***h*, **461** may include a parallel-connected combination of two or more discrete devices, such as IGBT devices.

(13) The liquid-cooled inverter **24** is shown as a three-phase device having three of the phase drivers **42***a*, **42***b*, **42***c*. However, the principles of the present disclosure may be implemented in a single-phase device having only one of the phase drivers **42***a*, **42***b*, **42***c* and/or in a multi-phase device having a different number, such as five or nine, of the phase drivers **42***a*, **42***b*, **42***c*.

(14) Each of the phase drivers **42***a*, **42***b*, **42***c* also includes two DC link capacitors **48** connected between the DC positive conductor **40***a* and the DC negative conductor **40***b* adjacent to the power switches **46***h*, **461** to supply relatively large inrush currents to the power switches **46***h*, **461** and to reduce electromagnetic interference (EMI).

(15) FIG. **3** illustrates a printed circuit board assembly (PCBA) **50** of the liquid-cooled inverter **24** in accordance with an aspect of the subject disclosure. The PCBA **50** may function as one of the phase drivers **42***a*, **42***b*, **42***c* of the liquid-cooled inverter **24**. The PCBA **50** includes a number of electric components necessary to convert DC current into AC current, such as switches, transistors, semiconductors and the like. As shown on FIG. **3**, the PCBA **50** includes a printed circuit board (PCB) **52** that defines a first surface **54** and a second surface **56** opposite the first surface **54**. The PCBA **50** includes three first switch devices **60** disposed on the first surface **54** of the PCB **52**. Each of the first switch devices **60** is a discrete device, such as single insulated gate bipolar transistor (IGBT) device. However, the first switch devices **60** may include other types of discrete electronic devices, such as a field-effect transistor or another type of junction device. Each of the first switch devices **60** includes a conductive tab **61** that is electrically connected to a drain terminal thereof, such as the collector C. The first switch devices **60** extend perpendicularly to the first surface **54** of the PCB **52** and are aligned in-line with one another, with the respective conductive

tabs **61** being co-planer with one-another. The first switch devices **60** may be electrically connected in parallel with one-another and together function as the high-side power switch **46***h*. The PCBA **50** also includes ten (10) of the DC link capacitors **48** disposed on the first surface **54** of the PCB **52**. However, the PCBA **50** may include a different number of the DC link capacitors **48**.
(16) The PCBA **50** may include three second switch devices **62** disposed on the second surface **56** of the PCB **52** and which are similar or identical to the first switch devices **60**. The second switch devices **62** may be arranged in-line with one-another defining a plane that is parallel to and spaced apart from the plane of the first switch devices **60**. The second switch devices **62** may be electrically connected in parallel with one-another and together function as the low-side power switch **461**. In some embodiments, one or more additional components, such as a gate driver for supplying control signals to the switch devices **60**, **62**, may be disposed on the PCB **52** of the PCBA **50**.
(17) The liquid-cooled inverter **24** may have a power rating that it can be met with discrete devices, such as a parallel combination of the first switch devices **60** and/or the second switch devices **62**, in contrast to a single power module common in inverters configured for higher power levels. Each of the power switches **46***h*, **461** of the liquid-cooled inverter **24** of the present disclosure includes three discrete switch devices **60**, **62** connected in parallel. Consequently, the total number of discrete switch devices **60**, **62** for the liquid-cooled inverter **24** is eighteen (i.e. three first switch devices **60** in each high-side power switch **46***h* and three second switch devices **62** in each low-side power switch **461** for each of three phases a,b,c). Each of the discrete switch devices **60**, **62** may require additional cooling surface area to maintain junction temperatures within operating ranges specified by a supplier or manufacturer thereof.
(18) FIG. **4** shows a top view of the PCBA **50** with a first heatsink **70** connected to the first switch devices **60**. The first heatsink **70** may provide rigidity to physically support the first switch devices **60**. The first heatsink **70** may be configured to carry electrical current. For example, the first heatsink **70** may include a metal material, such as copper, which may be electrically connected to the conductive tabs **61** of each of the first switch devices **60**. The first heatsink **70** may function as a bus bar to distribute electrical current between several of the first switch devices **60**. For example, the first heatsink **70** may function as the DC positive conductor **40***a*. The first heatsink **70** overlies and connects to the conductive tabs **61** of each of the first switch devices **60**, thereby reducing electromagnetic interference (EMI) that may be otherwise produced by operation of the first switch devices **60**.
(19) Thus, as illustrated in FIG. **4**, the liquid-cooled inverter **24** includes at least one first heatsink **70** which is attached to the circuit board to add rigidity and to carry current. The first heatsink **70** may provide thermal dissipation by transmitting heat away from the first switch devices. The first heatsink **70** may also provide structural rigidity against shock and vibration. In accordance with the disclosure, the entire contents of the liquid-cooled inverter **24**, including the PCBA **50**, may be configured to be immersed in flowing oil, which will cool the heatsink first heatsink **70**.
(20) FIG. **5** shows a perspective view of the liquid-cooled inverter **24**, including a stack of three of the PCBAs **50**, with each of the PCBAs **50** implementing a corresponding one of the phase drivers **42***a*, **42***b*, **42***c*. Only one of the PCBAs **50** associated with the A-phase driver **42***a* is detailed in FIG. **5**. However, each of the PCBAs **50** may have a similar or identical construction. FIG. **5** also shows one of the second switch devices **62** disposed on the second surface **56** of one of the PCBs **52**. FIG. **5** also shows a second heatsink **72** that is mechanically connected to the plurality of second switch devices **62** and configured to remove heat therefrom.
(21) In some embodiments, the second switch devices **62** that comprise one or more of the low-side power switches **461** may be electrically isolated from the second heatsink. For example, a sheet of material that is an electrical insulator and a good thermal conductor, such as a ceramic, may be disposed between the conductive tab **61** of one or more of the second switch devices **62** and the second heatsink **72**. This may be necessary where the second heatsink **72** is mechanically

connected to two or more discrete devices that are not connected in parallel. For example, in a case where the second switch devices **62** are associated the low-side power switches **461** of two or more of the phase drivers **42**_a_, **42**_b_, **42**_c_ and are mechanically connected to a same a second heatsink **72**. The second heatsink **72** may, therefore, provide mechanical stability to second switch devices **62** and to any other devices attached thereto.

(22) Alternatively, the second heatsink **72** may be electrically connected to the conductive tabs **61** of each of the second switch devices **62** where all of the second switch devices **62** mechanically connected thereto are connected in parallel. The second heatsink **72** may function as a bus bar to distribute electrical current between several of the second switch devices **62**. For example, where two or more of the second switch devices **62** are associated with the low-side power switch **461** of a given one of the phase drivers **42**_a_, **42**_b_, **42**_c_, the second heatsink **72** may be electrically connected thereto and function as the corresponding one of the output conductors **44**_a_, **44**_b_, **44**_c_. Thus, in some embodiments, the second heatsink **72** may function as a current carrying conductor.

(23) As shown in FIG. **5**, the first heatsink **70** includes fins **76** configured to increase surface area and to facilitate heat transfer to the liquid coolant. The second heatsinks **72** may include similar fins **76**. It should be appreciated that the fins **76** may have a different arrangement than what is shown on FIG. **5**, such as a different orientation and/or location. In some embodiments, the first heatsinks **70** and/or the second heatsinks **72** may define flow channels (not shown in the FIGS.), and which are configured to transmit the liquid coolant therethrough. Each of the first heatsinks **70** and the second heatsinks **72** may have a design that it is optimized for both heat transfer and for reduction of EMI.

(24) FIG. **5** also shows a rectangle that schematically represents a housing **80** that defines the inverter passageway **26** and which surrounds each of the PCBAs **50** of the liquid-cooled inverter **24**. The inverter passageway **26** may be configured as one or more chambers that hold the PCBAs **50** and which conduct a liquid coolant, such as the lubricant/coolant fluid **18**, in thermal communication with the first heatsinks **70** with the second heatsinks **72** for removing heat therefrom. The housing **80** may be made of a material, such as resin or metal, that is thermally insulating, and which reduces overall EMI. In some embodiments, the housing **80** may be integrally packaged with the lubricant-supported electric motor **11**. In some embodiments, the liquid-cooled inverter **24** may be configured such that the PCBs **52** are oriented vertically when the liquid-cooled inverter **24** is installed for operation, such as in a vehicle. Such vertical orientation may reduce or eliminate foaming in the liquid coolant therein, which could otherwise adversely affect efficiency of heat transfer between the heatsinks **70**, **72** and the liquid coolant.

(25) In some embodiments, the housing **80** of the liquid-cooled inverter **24** may include and/or function as the reservoir **38**, thereby alleviating need for a separate vessel.

(26) In some embodiments, the liquid-cooled inverter **24** may include a controller board (not shown in the FIGs.), which is disposed on a lid of the housing **80** and which electrically connects with each of the PCBAs **50**, e.g. using one or more pin and plug connectors. Such configuration may avoid an additional connector and allow easier replacement/servicing;

(27) In some embodiments, the inverter passageway **26** may be configured to cause the liquid coolant to have a turbulent flow, thereby increasing heat transfer from the first heatsinks **70** and/or the second heatsinks **72** and into the liquid coolant.

(28) In some embodiments, the PCBAs **50** of the liquid-cooled inverter **24** are connected together by one or more structures (not shown in the FIGS), such as stand-offs and/or by mounting tabs integrated in the housing **80**, and which maintain spacing between the respective PCBAs **50**.

(29) The inverter passageway **26** may be configured to distribute the liquid coolant and to maintain temperatures of the DC link capacitors **48** and the switch devices **60**, **62** of the power switches **46**_h_, **461** within predetermined acceptable temperature ranges.

(30) A common lubrication and cooling system includes a lubricant-supported electric motor including a stator and a rotor defining a gap therebetween, with a liquid coolant disposed in the gap

for supporting the rotor while allowing the rotor to rotate relative to the stator. The common lubrication and cooling system also includes a liquid-cooled inverter. The liquid-cooled inverter includes a plurality of switch devices configured to supply an alternating current (AC) power to the lubricant-supported electric motor for driving the rotor to rotate. The liquid-cooled inverter also includes a first heatsink mechanically connected to the plurality of switch devices. The liquid-cooled inverter also includes an inverter passageway configured transmit the liquid coolant between the lubricant-supported electric motor and into fluid communication with the first heatsink for transmitting heat away from the first heatsink.

(31) In some embodiments, the common lubrication and cooling system further includes a pump configured to circulate the liquid coolant between the lubricant-supported electric motor and through the inverter passageway of the liquid-cooled inverter.

(32) In some embodiments, the first heatsink is configured to conduct electrical current with the plurality of switch devices mechanically connected thereto.

(33) In some embodiments, the liquid-cooled inverter further comprises a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween. In some embodiments the first heatsink is electrically connected to the DC positive conductor, and each switch device of the plurality of switch devices includes a tab in electrical communication with the first heatsink.

(34) In some embodiments, the liquid-cooled inverter further comprises: a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween, and a phase driver. In some embodiments, the phase driver includes a printed circuit board (PCB), a high-side power switch configured to selectively conduct power between the DC positive conductor and an output conductor, and a low-side power switch configured to selectively conduct power between the DC negative conductor and the output conductor. In some embodiments, at least one of the high-side power switch and the low-side power switch includes the plurality of switch devices being disposed on a first surface of the PCB.

(35) In some embodiments, the high-side power switch comprises the plurality of switch devices. In some embodiments, the liquid-cooled inverter further comprises: the low-side power switch including a plurality of second switch devices disposed on the PCB, and a second heatsink mechanically connected to the plurality of second switch devices and configured to remove heat therefrom.

(36) In some embodiments, the plurality of second switch devices of the low-side power switch are located on a second surface of the PCB opposite from the first surface of the PCB with the plurality of switch devices of the high-side power switch

(37) In some embodiments, the plurality of second switch devices of the low-side power switch are electrically isolated from the second heatsink.

(38) In some embodiments, the phase driver is one of a plurality of phase drivers of the liquid-cooled inverter, with each phase driver of the plurality of phase drivers having a similar construction. In some embodiments, the PCBs of the plurality of phase drivers are stacked parallel to and spaced apart from one another.

(39) In some embodiments, the PCB and the first heatsink are submerged in the liquid coolant.

(40) In some embodiments, the plurality of switch devices of the at least one of the high-side power switch and the low-side power switch includes three of the switch devices.

(41) A liquid-cooled inverter includes a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween. The liquid-cooled inverter also includes a phase driver. The phase driver includes a printed circuit board (PCB), a high-side power switch configured to selectively conduct power between the DC positive conductor and an output conductor, and a low-side power switch configured to selectively conduct power between the DC negative conductor and the output conductor. At least one of the high-side power switch and the low-side power switch includes a plurality of switch devices disposed on the PCB. The liquid-

cooled inverter also includes a first heatsink mechanically connected to the plurality of switch devices and configured to remove heat therefrom. The liquid-cooled inverter also includes a housing defining an inverter passageway configured to conduct a liquid coolant in thermal communication with the first heatsink for removing heat therefrom.

(42) In some embodiments, the high-side power switch comprises the plurality of switch devices. In some embodiments, the liquid-cooled inverter further comprises: the low-side power switch including a plurality of second switch devices disposed on the PCB, and a second heatsink mechanically connected to the plurality of second switch devices and configured to remove heat therefrom.

(43) In some embodiments, the plurality of switch devices of the high-side power switch are disposed on a first surface of the PCB. In some embodiments, the plurality of second switch devices of the low-side power switch are located on a second surface of the PCB opposite from the first surface of the PCB with the plurality of switch devices of the high-side power switch.

(44) In some embodiments, the plurality of second switch devices of the low-side power switch are electrically isolated from the second heatsink.

(45) In some embodiments, the phase driver is one of a plurality of phase drivers each having a similar construction. In some embodiments, the PCBs of the plurality of phase drivers are stacked parallel to and spaced apart from one another.

(46) In some embodiments, the PCB and the first heatsink are submerged in the liquid coolant.

(47) In some embodiments, the first heatsink is configured to conduct electrical current with the plurality of switch devices mechanically connected thereto.

(48) In some embodiments, the first heatsink is electrically connected to the DC positive conductor. In some embodiments, each switch device of the plurality of switch devices includes a tab in electrical communication with the first heatsink.

(49) In some embodiments, the plurality of switch devices of the at least one of the high-side power switch and the low-side power switch includes three of the switch devices

(50) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A common lubrication and cooling system, comprising: a lubricant-supported electric motor including a stator and a rotor defining a gap therebetween, with a liquid coolant disposed in the gap for supporting the rotor while allowing the rotor to rotate relative to the stator; and a liquid-cooled inverter including: a plurality of switch devices configured to supply an alternating current (AC) power to the lubricant-supported electric motor for driving the rotor to rotate; a first heatsink mechanically connected to the plurality of switch devices; a phase driver including a printed circuit board (PCB); and an inverter passageway configured to transmit the liquid coolant between the lubricant-supported electric motor and into completely submerged relationship with the first heatsink and the PCB for liquid cooling the liquid-cooled inverter.

2. The system of claim 1, further comprising a pump configured to circulate the liquid coolant between the lubricant-supported electric motor and through the inverter passageway of the liquid-cooled inverter.

3. The system of claim 1, wherein the first heatsink is configured to conduct electrical current with the plurality of switch devices mechanically connected thereto.

4. The system of claim 1, wherein the liquid-cooled inverter further comprises a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween; wherein the first heatsink is electrically connected to the DC positive conductor, and wherein each switch device of the plurality of switch devices includes a tab in electrical communication with the first heatsink.
5. The system of claim 1, wherein the liquid-cooled inverter further comprises: a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween; and the phase driver further including: a high-side power switch configured to selectively conduct power between the DC positive conductor and an output conductor, and a low-side power switch configured to selectively conduct power between the DC negative conductor and the output conductor, wherein at least one of the high-side power switch and the low-side power switch includes the plurality of switch devices being disposed on a first surface of the PCB.
6. The system of claim 5, wherein the high-side power switch comprises the plurality of switch devices, and wherein the liquid-cooled inverter further comprises: the low-side power switch including a plurality of second switch devices disposed on the PCB; and a second heatsink mechanically connected to the plurality of second switch devices and configured to remove heat therefrom.
7. The system of claim 6, wherein the plurality of second switch devices of the low-side power switch are located on a second surface of the PCB opposite from the first surface of the PCB with the plurality of switch devices of the high-side power switch.
8. The system of claim 6, wherein the plurality of second switch devices of the low-side power switch are electrically isolated from the second heatsink.
9. The system of claim 5, wherein the phase driver is one of a plurality of phase drivers of the liquid-cooled inverter, with each phase driver of the plurality of phase drivers having a similar construction; and wherein the PCBs of the plurality of phase drivers are stacked parallel to and spaced apart from one another.
10. The system of claim 5, wherein the plurality of switch devices of the at least one of the high-side power switch and the low-side power switch includes three of the switch devices.
11. A liquid-cooled inverter, comprising: a direct current (DC) positive conductor and a DC negative conductor configured to have a DC voltage therebetween; a phase driver including: a printed circuit board (PCB), a high-side power switch configured to selectively conduct power between the DC positive conductor and an output conductor, and a low-side power switch configured to selectively conduct power between the DC negative conductor and the output conductor, and wherein at least one of the high-side power switch and the low-side power switch includes a plurality of switch devices disposed on the PCB; a first heatsink mechanically connected to the plurality of switch devices and configured to remove heat therefrom; and a housing defining an inverter passageway configured to conduct a liquid coolant into completely submerged relationship with the first heatsink and the PCB for liquid cooling the liquid cooled inverter.
12. The liquid-cooled inverter of claim 11, wherein the high-side power switch comprises the plurality of switch devices, and wherein the liquid-cooled inverter further comprises: the low-side power switch including a plurality of second switch devices disposed on the PCB; and a second heatsink mechanically connected to the plurality of second switch devices and configured to remove heat therefrom.
13. The liquid-cooled inverter of claim 12, wherein the plurality of switch devices of the high-side power switch are disposed on a first surface of the PCB; and wherein the plurality of second switch devices of the low-side power switch are located on a second surface of the PCB opposite from the first surface of the PCB with the plurality of switch devices of the high-side power switch.
14. The liquid-cooled inverter of claim 12, wherein the plurality of second switch devices of the low-side power switch are electrically isolated from the second heatsink.
15. The liquid-cooled inverter of claim 11, wherein the phase driver is one of a plurality of phase

drivers each having a similar construction; and wherein the PCBs of the plurality of phase drivers are stacked parallel to and spaced apart from one another.

16. The liquid-cooled inverter of claim 11, wherein the first heatsink is configured to conduct electrical current with the plurality of switch devices mechanically connected thereto.

17. The liquid-cooled inverter of claim 16, wherein the first heatsink is electrically connected to the DC positive conductor, and wherein each switch device of the plurality of switch devices includes a tab in electrical communication with the first heatsink.

18. The liquid-cooled inverter of claim 11, wherein the plurality of switch devices of the at least one of the high-side power switch and the low-side power switch includes three of the switch devices.